

AI See, You See: Human-AI Musical Collaboration in Augmented Reality

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Figure 1: Left: Using our co-creative system for trio performances. Right: A view through the AR headset showing real-time visualisations of both the human collaborator’s actions (left keyboard) and an AI musician’s actions (right keyboard).

Abstract

The integration of artificial intelligence (AI) into creative practices has opened new possibilities for dynamic human-AI collaboration. While prior research has explored structured, process-based co-creative tasks, real-time, open-ended creative contexts such as musical performance remain underexplored. These environments demand tools that facilitate dynamic interaction, mutual understanding, and the explainability of AI contributions. This work introduces a novel augmented reality (AR) co-creative musical system that visualises AI’s musical actions, such as MIDI outputs and control parameters, through real-time AR visualisations. By embedding these visualisations within the performance space, the system facilitates shared creative processes and elevates AI’s presence. Using an auto-ethnographic approach, we evaluated the system in three musical performances across varied configurations, revealing insights into the role of AR in shaping human-AI collaboration and co-creative dynamics. This work offers design and evaluation insights for AR-enabled co-creative systems.

CCS Concepts

• **Applied computing** → **Sound and music computing**; • **Human-centered computing** → *Empirical studies in collaborative and social computing*; **Mixed / augmented reality**.

Keywords

Human-AI Collaboration, Augmented Reality (AR), Digital Musical Instruments (DMIs)

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1 Introduction

The integration of artificial intelligence (AI) in creative practices has opened new opportunities for dynamic collaboration between humans and machines. However, despite technical advancements in intelligent systems [35], their application in real-time creative contexts remains an emerging research area [23, 27, 46]. A primary challenge is enabling humans to effectively understand, interpret, and engage with AI contributions. In this context, explainability and sense-making of AI’s actions are essential for fostering trust [33, 41], facilitating collaboration [9], and enhancing the creative process [5]. Real-time creative collaboration, such as musical performance, requires participants to respond dynamically to evolving inputs and contexts [8, 30, 43]. Co-creative systems must exhibit a level of agency that allows them to interact meaningfully with human

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collaborators, adapting to their actions and facilitating a shared creative vision [9, 27]. While prior research has explored structured, process-based co-creative tasks like drawing [9, 15] and music composition [37], this work seeks to explore the under-explored demands of real time, interactive contexts for tools that facilitate engagement and mutual understanding.

Artistic HCI research emphasises embodied and performative approaches, where physical interaction and sensory feedback produce knowledge through practice [14, 20, 39] and provides a rich lens for understanding human-AI collaboration. Unlike task-oriented activities, artistic performance foregrounds emotional and enactive experiences, which are central to the open-ended, improvisational nature of co-creative processes [19, 26]. Augmented reality (AR) offers promising opportunities to enhance performative and interactive dimensions of co-creation. Unlike traditional visualisation approaches such as static or abstract representations, AR interfaces provide spatial and immersive feedback of musical information and process [1, 13, 25]. This enables them to visualise AI actions in ways that are intuitive, context-sensitive, and directly aligned with the creative environment [40], while supporting real-time, spatially aligned feedback that enhancing mutual understanding and deepening the collaborative relationship between human and AI agents.

In this work, we present an AR co-creative musical system designed to support real-time interaction and collaboration between a human musician and an AI agent. The system visualises the AI's musical actions—such as MIDI outputs including musical keys and control parameters—through an AR interface, allowing performers to perceive and respond to the AI's contributions more effectively. By embedding this visual feedback directly within the performance space, the system supports an engaging and interactive co-creative process. To explore the system's creative potential and flexibility, we report on our experiences in three musical performances using different instruments and configurations. These performances provide valuable insights into the dynamics of human-AI collaboration, revealing how AR-enabled visualisations affect human perceptions of AI contributions and the overall creative workflow. The key contributions of this work are:

- (1) A novel AR interface for co-creative musical systems: demonstrating how AR visualisations of AI actions support collaboration by providing spatially and contextually aligned feedback.
- (2) Empirical insights into AR interface design for co-creative systems: employing an autoethnographic approach to evaluate how AR visualisations shape the perception of AI's role, enhance co-creative dynamics, and mediate human-AI collaboration, while paving the way for future research on comprehensive system design.

2 Related Work

2.1 Performing with AI in Music and Creative Arts

The intersection of AI and creative arts has gained increasing attention. While much of the existing research emphasises the technical

complexity and capabilities of intelligent systems [46], a growing body of work explores their role in dynamic, co-creative processes [23, 27]. Existing research categorised AI's role in co-creative musical systems into three primary types [11]: a technical tool for processing and analysing musical information, a creativity support tool, and an active collaborator [4, 27]. A co-creative model which both AI and human participate has been applied to practices such as sound design [12, 31], musical robotics [41] and live coding [44, 45].

Human-AI musical collaboration further explores the evolving relationships between musicians and AI agents. Vear et al. [41] examined an inclusive musical AI robot, uncovering varied perceptions of the role of an AI collaborator. The study revealed that AI could act as an ambiguous collaborator provoking imagination, or as a non-judgemental accompanist stimulating creativity through group dynamics. While these findings highlighted AI's potential to engage with musicianship towards a shared creative goal, they also underscored the challenges of establishing common ground in human-machine relationships, including embracing risks and aesthetic failures. Similar dynamics have been observed in other creative domains, such as dance [3, 39]. Wallace noted how glitches in AI systems can act as productive catalysts for co-creation by introducing unexpected disruptions that inspire novel creative directions [3]. However, these moments of productive ambiguity also raise questions about how artists interpret and respond to AI actions. Bryan-Kinns highlighted the need for explainability in creative AI systems, questioning how AI actions might be interpreted [5] in real-time feedback loops. Our work seeks to address these challenges by using AR visualisations to enhance the explainability of real-time AI actions in human-AI musical performances.

2.2 Augmented Reality, Communication and Collaboration

AR offers unique advantages as a visualisation modality in collaborative systems. By enabling the spatial arrangement of virtual content, AR facilitates the organisation and structure of complex ideas, supporting sense-making and communication among collaborators [16, 29]. For instance, Luo et al. demonstrated that physical layouts of documents in AR could act as an abstraction of users' thoughts and strategies, reflecting their processes of thinking and externalisation [16]. This highlights opportunities for AR to support spatially flexible visualisations that enhance communication and the presentation of sense-making outcomes. Rezewana et al. [27] emphasised the importance of supporting sense-making in human-AI co-creative systems, identifying the challenge of establishing common ground for communication between human and machine to coordinate each other's musical decisions in collaboration [7]—a problem often drawn from human-human collaboration. AR's ability to provide real-time, spatially aligned feedback related to physical environments offers a potential solution by improving mutual understanding in dynamic and creative workflows. In musical contexts, AR has been employed to project real-time visualisations directly onto physical instruments [1], supporting communication of musical processes. Other applications include 3D visualisations to enhance spatial audio experiences and playful engagement in collaborative music-making [13, 25]. Other creative applications such as in dance and theatre have used AR to visualise dancers'

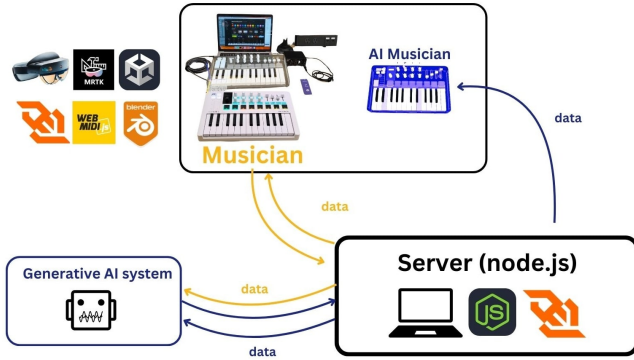


Figure 2: A system digram showing our human-AI co-creative system. The MIDI interaction data is sent over a network to enable real-time communication between human and AI musicians.

3D movements to support both artistic expression and coordination [6, 42]

Existing visualisation approaches in co-creative systems often rely on static or less immersive representations. Examples include visual representations of musical notes [26], AI confidence levels and human musicians’ physiological data (e.g., skin conductance) [19], or drawing and sketches [9, 15], to convey collaborators’ actions and intentions. While such tools effectively support real-time feedback, they lack the spatial and immersive qualities that AR uniquely offers. The integration of AR into human-AI co-creative systems could bridge this gap by providing more intuitive and context-sensitive visualisations. In other domains, such as human-robot interaction, AR visualisations have been leveraged to bridge communication and decision making [36]. Examples include 3D animations and prescriptive guidance to present proactive information about upcoming actions or plans. These applications illustrate the potential for AR to improve explainability, communication, and collaboration in human-AI co-creative systems.

Despite its potential, AR interfaces remain underexplored in intelligent musical systems. Turchet et al. highlighted the benefits of extended reality (including AR) in these contexts for spatial representation and flexibility [40]. These features could enable real-time, context-sensitive feedback on AI actions, aligning with the need for tailored explanations in explainable AI to support dynamic creative processes [5]. Our work leverages an AR interface that presents MIDI information, a widely used musical protocol in intelligent musical systems [38]. By visualising the AI agent’s real-time musical activities, we aim to explore how AR affects human-AI co-creation dynamics and collaboration quality.

3 System Design

The co-creative system, as shown in Figure 2, consists of two main components: the generative AI system, and the AR performer interface. These components are connected through a Node.js [24] WebSocket server, enabling real-time communication. Musical interaction (MIDI) data, captured from the AI system (representing AI’s action), are transmitted as WebSocket messages and

visualised on the AR interface in real-time [10]. The human musicians connect their electronic musical instrument’s MIDI outputs to the generative AI system’s MIDI inputs, allowing their musical actions to directly influence the AI’s responses. The following sections describe each component.

3.1 Generative AI System

We applied the open-source IMPSy system [17, 18] as a back-end AI engine for experimental computer music. IMPSy is an artist-centred system for incorporating a generative AI model into real-time musical interactions. This system processes MIDI input from human musicians and generates responsive musical outputs through sound played on a connected instrument. Human performers connect their MIDI-compatible instruments (e.g., synthesisers, keyboards) to IMPSy’s MIDI input. The system analyses this input in real time to inform its musical predictions and outputs, which are simultaneously visualised on the AR performer interface. This generative AI system uses multiple MIDI channels to encode its musical actions, including pitch, velocity, and control changes (e.g., timbral adjustments or effect knobs). These outputs are mapped to corresponding MIDI-compatible devices or software synthesisers, allowing the AI-generated music to be played in real-time in our performance setup. IMPSy was chosen over other generative AI music systems such as Notochord [32] and Magenta Studio [28] due to the capacity to control multiple synthesis parameters through MIDI and to exchange information through both MIDI and WebSocket connections.

3.2 AR performer interface

The AR interface is a 3D hologram displayed via a Microsoft HoloLens 2 AR headset [21], and our system was developed in Unity (v2020.3.5f1) with Microsoft Mixed Reality Toolkit [22] (v2.8.3). The interface is modelled on a Arturia MiniLab 3 electronic music keyboard, similar to many widely-available small electronic keyboards featuring a two-octave piano-style keyboard, faders, pads, and encoders. Our choice of this interface is to provide a generic representation of a digital musical instrument, allowing musicians to identify which musical keys or knobs collaborators were interacting with, thereby aiding coordination and communication of musical ideas.

The interface visualises ongoing MIDI events generated by the AI system sent over the network, ensuring compatibility with a wide range of electronic synthesisers and MIDI devices. When the AI outputs a musical note, the corresponding key on the virtual keyboard lights up in a yellow colour, indicating pitch and timing; while these cues are audible, the AR interface supplements them with immediate visual feedback that may reduce cognitive load and improve coordination in complex performances. Similarly, the seven effect knobs on the interface dynamically reflect the AI’s adjustments to parameters such as filters, modulation, and resonance, offering real-time insights into its creative contributions. This multimodal feedback is intended to allow human musicians to interpret the AI’s contributions more effectively, fostering clearer communication and enhancing collaborative creativity. For this study, the interface was tailored to display the musical keys and seven effect knobs, mapping them directly to the AI system’s MIDI output channels.

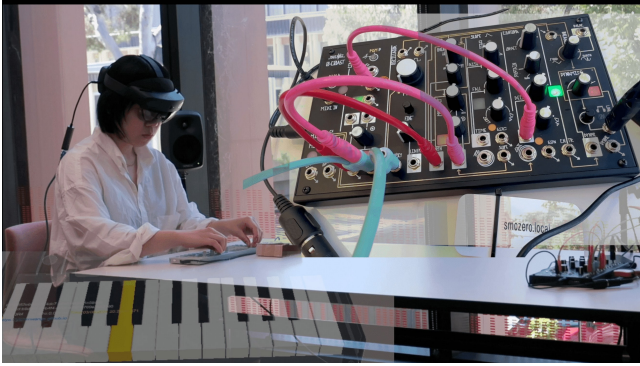


Figure 3: The performer operates an OP-1 synthesiser while the AI system controls a 0-COAST synthesiser (right). The overlay in the bottom-left corner displays the keys being played by the AI, as viewed through the AR headset.

4 Experiences and Performances

We have utilised our AR co-creative musical system in several musical performances, gathering insights and reflections to iteratively update the system along the way. Both authors have extensive experience in developing and performing with computer music systems that integrate novel technologies (e.g., AR, iPad instruments, intelligent musical systems). Our backgrounds in experimental, open-ended performance practices provided critical insights during these sessions. We report on three experiences in order.

4.1 Recorded Demo Performance

The first formal solo performance using the system was recorded in February 2024 with a 0-Coast synthesiser. In this performance, the human performer played an OP-1 synthesiser, while the AI agent controlled a 0-Coast synthesiser, as shown in Figure 3. At this stage of the exploration, the AI agent was limited to a single channel of MIDI note output, allowing it control only pitch and not timbral changes on the 0-Coast. Three performances were recorded, each with an average length of 8 minutes. Although the sonic palette was limited, these recordings demonstrated the feasibility of visualising the AI agent’s actions through the AR performer interface. The human performer found the co-creative experience exciting, as the clear visualisation of the AI’s actions provided a better understanding of its contributions. During the performance, the human performer played keys and adjusted effects knobs to create a varied soundscape, incorporating different instrument styles such as percussive, melodic, and electronic, with some different musical patterns. The musician would see the visualisation of how AI played musical keys, but it played in such a quick way that was hard to follow.

The AI agent primarily maintained a drone-based texture with limited sonic variation. This highlights a significant challenge: the single MIDI channel restriction limited the AI agent’s creative potential. From this experience, we identified that the 0-Coast synthesiser’s limited MIDI control capabilities made it less suitable for this system. This insight suggests that exposing more parameters and controls for the human musician could enhance the co-creative



Figure 4: The performer played an OP-1 synthesiser during the concert, using a co-creative system to observe and respond to the AI’s musical actions through visualisations displayed in an AR headset.

interaction. A more capable synthesiser with broader MIDI support would allow the AI agent to explore a greater sonic variety, thereby enriching the collaborative experience.

4.2 Research Concert

In May 2024, our co-creative system was trialled in a live research-focussed concert as part of a program of works featuring computer music and AI. For this performance, the human used the same OP-1 synthesiser, while the AI agent performed with a Korg Volca FM synthesiser. The AI system was upgraded to support 8 MIDI inputs and outputs, allowing flexible mapping of sound parameters between the human musician and AI incorporations through synthesisers. This upgrade enabled greater timbral variation and provided the human performer with more creative options to achieve diverse artistic goals.

The human musician found the co-creation experience in this performance significantly more enjoyable and rewarding compared to previous performances. While the AI agent continued to generate percussive textures, more evident timbral changes were observed. Instead of focusing primarily on pitch differences, the AI-produced showcased greater variety in texture and resonance. This was attributed to the expanded control parameters of the FM synthesiser, which the AI agent operated. Timbral parameters, such as the “LFO” and “algorithm” settings, introduced additional dimensions of sonic variability. The interaction between the human and the AI showed dynamic patterns of call and response. The human musician, for example, could play a melodic pattern, adjust effect knobs, and then observe the AI’s reaction through both visual feedback in the AR interface and corresponding changes in sound. This interplay created several memorable moments during the 9-minute performance, and the musician noted that they could have comfortably continued playing for a longer duration.

Reflecting on this performance, we observed that switching to a different synthesiser did not necessarily alter the human musician’s creative process or specific musical choices. Instead, it was the

expanded control options provided by the co-creative AI system that encouraged the musician to explore new ways of influencing the AI's behaviour.

4.3 Studio Trio Improvisation

In January 2025, the co-creative system was tested in a trio improvisation setting, featuring two human musicians and the AI agent. Building on insights from previous performances, this session investigated whether the perception and roles of the AI would evolve with the support of an AR interface, and how does it affect the shared understanding.

Two Arturia MiniLab 3 controllers were used by the human musicians with sounds produced by Arturia Analogue Lab V [2] software synth library. Both the human and AI musicians could assign different sounds (e.g., bass, lead, pads, drums, sound effects) in their instrument setup. The AR interface was adjusted to visualise all eight MIDI outputs of the AI system, including key presses and 7 effect knobs, and include two performer interface visualisations, displaying the musical actions of both the AI and the second human musician, as shown in Figure 1. Each human musician's MIDI inputs were sent to the AI system via the WebSocket network, and their sonic outputs were visualised on the AR interface. The AI's musical actions were visualised on the right interface, while the left visualisation represented the human collaborators' actions, offering a comprehensive view of the performance dynamics. In these trio improvisation sessions, two topologies were explored:

- T1:** Shared AI Instrument: A musician and the AI shared a single instrument.
- T2:** Independent AI Instrument: The AI operated a separate instrument alongside the human musicians.

Three performances were recorded with an average length of 8 minutes. After the last recording session, the authors discussed their co-creation experiences and the AR visualisations in shaping mutual understanding during the process. The conversation were audibly recorded and transcribed in Aiko [34]. We summarise the key reflections as follows.

First, we found the topology setup influenced how the musicians perceived and interacted with the AI. Musician YW (first author) found the **T1** setup simpler and viewed AI as an accompanist, on which she musically focused to keep track of the AI's playing while trusting Musician CM (second author) to play an atmospheric texture. Musician CM had different views. He perceived the AI as a distinct performer with more autonomous contributions, comparable to a human collaborator: *"an AI performer [should be] on a separate instrument."* Both musicians agreed that separating the AI onto its own instrument clarified its contributions and emphasised its individuality within the ensemble.

Regarding visualisations, both musicians found it served as a critical reference for understanding and responding to both the AI and the humans' actions, working towards the collaborative goal. Musician YW particularly engaged with the AR performer interface to explore musical ideas, but found it challenging to follow due to how fast the AI played musical keys. In contrast, the human performer's visualisation was much easier to interpret and respond to. For Musician CM, he noted that the visualisation revealed both the AI's strengths and its limitations (e.g., identifying when its actions

were less meaningful or relevant). This increased transparency presented a sense of in control helped shape their responses during the improvisation. Compared to other performance experiences with the AI system without a visual reference, where the musician has to listen and look at the other player and try to figure out what's going on. He found it quite novel to have a visualisation of what everyone was doing. More importantly, Musician CM commented that the AR visualisation of both AI and human performer elevated AI's presence, making it feel more like a fully engaged ensemble member rather than just a background element. It placed more emphasis on which note AI was actually playing, which could have less attention paid otherwise.

Both musicians acknowledged that the AR visualisation shifted their attention and how it affects their cognitive load during the performance. Musician CM found himself focusing less on their co-performer and more on the AR visual feedback of the musical information. Despite this shift, they did not perceive a negative impact on the performance, as auditory cues ensured sufficient awareness of the ensemble's overall sound.

With a more complex usages of the AR co-creative system, we found that AR visualisations mediated the understanding of the AI's contributions and social dynamics in the creative collaboration. The visualisations proved valuable for fostering mutual understanding and elevating the AI's perceived presence in the improvisation. Musicians could see the specific notes or changes made by the AI, prompting them to engage more thoughtfully with its contributions. An increased focus on the AR interface also underscored the trade-off in attention between visual and auditory cues during co-creation.

5 Conclusion and Future Work

In this work, we presented a novel AR co-creative system designed to enable real-time creative collaboration by bridging explainability and mutual understanding. By leveraging AR visualisations, the system facilitated dynamic interactions and shared creative processes in musical performance. We evaluated the system through auto-ethnography, performing with it in three structured musical contexts that explored different configurations of the AI system and AR visualisations. Our performance reflections reveal that real-time AR visualisations play a crucial role in supporting human musicians' understanding of AI contributions. By providing immersive, real-time representations of AI actions, the AR interface enhanced an understanding of the musical process and enabled musicians to engage more effectively towards creative goals. For instance, richer visual feedback of musical keys and control parameters clarified AI contributions and elevated its role from a supporting tool to that of a distinct collaborator. Furthermore, the trio improvisations highlighted how the topology of shared versus independent AI setups influenced perceptions of the AI's agency, with shared setups emphasising its role as an accompanist and independent setups allowing it to provide more creative contributions. These findings suggest that increased visibility of AI's action encourages experimentation, enabling shifts in creative dynamics and interactions, such as musical choices and attention redirection.

While AR enhanced co-creation in musical contexts, it also introduced challenges related to cognitive load and attention management. Musicians in the trio improvisations reported that the

visualisations shifted their focus from auditory to visual feedback. Although this did not negatively affect performance flow, it underscores the need for balanced visualisation designs that integrate with auditory cues. Future iterations of the system should address these trade-offs to maintain collaborative flow while minimising cognitive demands on performers.

The insights from this study highlight several directions for future work. First, an in-depth investigation of one-on-one human-AI co-creation is needed to understand how AR visualisations influence collaboration dynamics including communication and mutual understanding. Building on this, expanding the scope to study human-AI team dynamics in group contexts will enable a comprehensive exploration of key aspects such as collaborative processes and the creative outcomes of these interactions. Finally, addressing the interplay between cognitive load, attention shifts, and performance flow is critical for designing AR interfaces that enhance co-creative workflows without overwhelming users. Additionally, future research should explore the broader user benefits and implications of AR-enhanced co-creative systems across various expertise levels and creative domains. By presenting these findings, this study lays the groundwork for leveraging AR in human-AI co-creative systems, offering valuable insights for the design and evaluation of tools that enhance collaboration in the creative arts. The AR co-creative system's ability to foster explainability and elevate the perceived agency of AI collaborators demonstrates its potential to transform creative practices, paving the way for more dynamic, interactive human-AI partnerships in music and beyond.

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